



iQ®200 Automated Urinalysis System

LIS Communication Information

1. Introduction

1.1. Purpose

This document provides information about communication between the Iris iQ®200 Automated Urinalysis System and the Laboratory Information System (LIS).

1.2. Definitions

LIS – Laboratory Information System is a software system that computerizes all aspects of record keeping and communication for a laboratory department.

ASTM – American Society for Testing and Materials.

Unicode – Binary codes setup for text or script characters. It is a system for the interchange, processing, and display of the written texts of the diverse languages.

XML – Extensible Markup Language.

iQ200 System – The Iris iQ®200 Automated Urinalysis System consists of a chemistry analyzer, AUTION MAX™ AX-4280 Automated Urine Chemistry Analyzer and a microscopy analyzer, iQ®200 Automated Urine Microscopy Analyzer.

1.3. Reference

ASTM 1381-95: Standard specification for low-level protocol to transfer messages between clinical laboratory instruments and computer systems.

2. Overview

2.1. Communications to LIS

- Unidirectional (Fig. 1)
- Bidirectional host query (Fig. 2)

LIS Interface

☒ Enable LIS

Communication settings

Bits per second: 9600

Data bits: 8

Parity: None

Stop bits: 1

Specimen information from LIS

☐ Obtain specimen information from LIS

☐ Skip specimen if LIS fails

☐ Obtain chemistry information from LIS

☐ Obtain patient demographics information from LIS

☐ Suppress usage of iQ® Series generated Chemistry Results when obtaining specimen information from LIS

Demographics...

Specimen information to LIS

☐ Suppress "[none]" results

☒ Suppress chemistry result that precedes consolidation with iQ® Series Microscopy Analyzer

OK Cancel

Fig. 1

LIS Interface

☒ Enable LIS

Communication settings

Bits per second: 9600

Data bits: 8

Parity: None

Stop bits: 1

Specimen information from LIS

☒ Obtain specimen information from LIS

☐ Skip specimen if LIS fails

☐ Obtain chemistry information from LIS

☐ Obtain patient demographics information from LIS

☐ Suppress usage of iQ® Series generated Chemistry Results when obtaining specimen information from LIS

Demographics...

Specimen information to LIS

☐ Suppress "[none]" results

☒ Suppress chemistry result that precedes consolidation with iQ® Series Microscopy Analyzer

OK Cancel

Fig. 2

3. Communication protocol

ASTM E1381-95 describes specifications for electronic transmission of messages between clinical laboratory instruments and computer systems.

3.1. Communication Phases

There are three distinct phases in transferring information between an instrument and a computer system. The three phases are establishment, transfer, and termination.

3.1.1. Establishment Phase

This is the phase where direction of information is determined. In establishing communication the sender transmits <ENQ> and will ignore responses other than <ACK>, <NAK> or <ENQ>.

3.1.2. Transfer Phase

During the transfer phase, the sender transmits messages to the receiver. The transfer phase ends when all messages are sent.

3.1.2.1. Message structure

ASTM specifies that when a message contains 240 characters or less, it is sent in a single end frame. Longer messages are sent in two or more frames.

The frame structure is listed below:

<STX> FN Text <ETB>/<ETX> C1 C2 <CR> <LF>

Where:

<STX> = Start of Text transmission control character

FN = Single digit Frame Number 0-7

Text = Data content of message

<ETB> = End of Transmission Block control character

<ETX> = End of Text transmission control character

C1 = Most significant character of checksum 0-9 and A-F

C2 = Least significant character of checksum 0-9 and A-F

<CR> = Carriage Return character

<LF> = Line Feed character

Data content format is ASCII encoded Unicode XML.

3.1.3. Termination Phase

In this phase the sender transmits <EOT> to inform the receiver that data transmission is terminated and regards the data link to be in neutral state.

4. Examples

This section shows examples of communications between the iQ200 System and the LIS.

4.1. Example 1:

The instrument sends <IRISPing/> to verify the communications link to the LIS.

```
<stx>13C004900520049005300500069006E0067002F003E000D000A00
<etx>C4<cr><lf>
```

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Data:

```
<IRISPing/>
```

4.2. Example 2:

The instrument sends <SIQ>URINSAMPLE_ID</SIQ> to query LIS, such as

Data:

```
<SIQ>323505316</SIQ>
```

4.3. Example 3:

LIS replies with Body Fluid type, WorkOrder and sample identifier.

Data:

```
<?xml version="1.0"?>
<SI BF="URN" WO="run">323505316</SI>
```

4.4. Example 4:

LIS replies with Body Fluid type, WorkOrder, sample identifier and patient demographics.

Data:

```
<?xml version="1.0" encoding="UTF-16"?>
<SI BF="URN" WO="run">1529934<DEMOG LName="Test" FName="Gregory" MName="P"
Loc="E/R" RecNum="" Gender="M" DOB="2001-01-01"/></SI>
```

4.5. Example 5:

LIS replies with Body Fluid type, WorkOrder and sample identifier. Below is example with chemistry results.

Data:

```
<?xml version="1.0" encoding="UTF-16"?>
<SI BF="URN" WO="run">323505316<RCS><RC Key="BLD">+1</RC><RC
Key="BIL">+2</RC><RC Key="URO">+3</RC><RC Key="KET">+4</RC><RC
Key="GLU">+</RC><RC Key="PRO">+4</RC><RC Key="NIT">-</RC><RC
Key="LEU">NEG.</RC><RC Key="PH">6.5</RC><RC Key="SG">1.015</RC><RC
Key="COL">Yellow</RC><RC Key="CLA">Clear</RC></RCS></SI>
```

4.6. Example 6:

LIS replies with Body Fluid type, WorkOrder, sample identifier and patient demographics. Below is example with chemistry results.

```

/LPF">[none]</AR><AR Key="CELL" SN="CELL" LN="Cellular Cast" AF="0" NR="1
/LPF">[none]</AR><AR Key="BROAD" SN="BROAD" LN="Broad Cast" AF="0" NR="1
/LPF">[none]</AR><AR Key="FATC" SN="FATC" LN="Fatty Cast" AF="0" NR="1 /LPF">[none]</AR><AR
Key="WAXY" SN="WAXY" LN="Waxy Cast" AF="0" NR="1 /LPF">[none]</AR><AR Key="UNCX"
SN="UNCX" LN="Unclassified Crystal" AF="0" NR="FEW">RARE</AR><AR Key="TPO4" SN="TPO4"
LN="Triphosphate Crystal" AF="0" NR="FEW">[none]</AR><AR Key="CAOX" SN="CAOX" LN="Calcium
Oxalate Crystal" AF="0" NR="FEW">[none]</AR><AR Key="CAPH" SN="CAPH" LN="Calcium Phosphate
Crystal" AF="0" NR="FEW">[none]</AR><AR Key="CACB" SN="CACB" LN="Calcium Carbonate Crystal"
AF="0" NR="FEW">[none]</AR><AR Key="URIC" SN="URIC" LN="Uric Acid Crystal" AF="0"
NR="FEW">[none]</AR><AR Key="LEUC" SN="LEUC" LN="Leucine Crystal" AF="0"
NR="POS">[none]</AR><AR Key="CYST" SN="CYST" LN="Cystine Crystal" AF="0"
NR="POS">[none]</AR><AR Key="TYRO" SN="TYRO" LN="Tyrosine Crystal" AF="0"
NR="POS">[none]</AR><AR Key="AMOR" SN="AMOR" LN="Amorphous Crystal" AF="0"
NR="FEW">[none]</AR><AR Key="UNCL" SN="UNCL" LN="Unclassified" AF="0" NR="99999999
/HPF">[none]</AR></AC><CM>Comment appears here!!!!!! MD
notified!!!!</CM><ARV>42</ARV><AUD><XE OP="GregLoa" dt="2006-08-30 11:29:02"/><XE
OP="GregLoa" dt="2006-08-29 16:51:21"/><XE OP="Iauto" dt="2006-08-29 16:42:32"/></AUD><SysCfg
MicroPresent="T" AXPresent="T" iChem200Present="F"/></SA>

```

4.8. Example 8:

Upon completion of analysis, instrument sends the following transmission (with patient demographics):

Data:

```

<?xml version="1.0" encoding="UTF-16"?>
<?xml-stylesheet type="text/xsl" href="C:\IRIS2K1\Templates\SpecimenAnalysis-style.xsl"?>
<SA BF="URN" ID="1529930" SID="AP" OP="GregLoa" ADT="2006-08-30 11:52:58-08:00" ADTS="2006-
08-30 11:52:58" RDT="2006-08-30 12:02:43-08:00" RDTS="2006-08-30 12:02:43" RP="4" SQN="521"
RSQN="5" DILN="1" DILD="1" IMP="0" LName="Test" FName="Gregory" MName="P" DOB="2001-01-01"
Loc="Outpatient" RecNum="20060830001" Gender="M" CDT="2006-08-30 11:52:58-08:00" CDTs="2006-
08-30 11:52:58" REDT="2006-08-30 12:02:43-08:00" REDTS="2006-08-30 12:02:43"><PF>Iris LIS
Laboratory</PF><PF>9172 Eton Ave</PF><PF>Chatsworth, CA
91311</PF><PF>800.776.4747</PF><PF></PF><PF></PF><PF></PF><AC AT="Chemistry" AS="Done"
SO="External" DSLOT="AX-Stick"><AR Key="GLU" SN="GLU" LN="Glucose" AF="0" NR="+-
">Normal</AR><AR Key="PRO" SN="PRO" LN="Protein" AF="0" NR="+- "></AR><AR Key="BIL" SN="BIL"
LN="Bilirubin" AF="0" NR="+1"></AR><AR Key="URO" SN="URO" LN="Urobilinogen" AF="0"
NR="+1">Normal</AR><AR Key="PH" SN="PH" LN="pH" AF="0" NR="[none]">6.0</AR><AR Key="BLD"
SN="BLD" LN="Blood" AF="0" NR="+- "></AR><AR Key="KET" SN="KET" LN="Ketone" AF="0" NR="+- ">
</AR><AR Key="NIT" SN="NIT" LN="Nitrite" AF="0" NR="+1"></AR><AR Key="LEU" SN="LEU"
LN="Leukocytes" AF="1" NR="25">75 Leu/ul</AR><AR Key="CLA" SN="CLA" LN="Clarity" AF="0"
NR="[none]">Clear</AR><AR Key="SG" SN="SPGR" LN="Specific Gravity" AF="0"
NR="[none]">1.004</AR><AR Key="COL" SN="COL" LN="Color" AF="0" NR="[none]">Light-
Yellow</AR></AC><AC AT="Sediment" AS="Done" SO="Internal"><AR Key="ART" SN="ART"
LN="Artifact" AF="0" NR="99999999 /HPF">[none]</AR><AR Key="RBC" SN="RBC" LN="Red Blood Cell"
AF="1" NR="3 /HPF">152 /HPF</AR><AR Key="DRBC" SN="DRBC" LN="Dysmorphic Red Blood Cell"
AF="0" NR="[none]">[none]</AR><AR Key="WBC" SN="WBC" LN="White Blood Cell" AF="0" NR="5
/HPF">[none]</AR><AR Key="WBCC" SN="WBCC" LN="White Blood Cell Clump" AF="0"
NR="RARE">[none]</AR><AR Key="BACT" SN="BACT" LN="Bacteria" AF="0"
NR="Few">[none]</AR><AR Key="BYST" SN="BYST" LN="Budding Yeast" AF="0"
NR="Rare">[none]</AR><AR Key="HYST" SN="HYST" LN="Hyphae Yeast" AF="0"
NR="Rare">[none]</AR><AR Key="SQEP" SN="SQEP" LN="Squamous Epithelial" AF="0" NR="5
/LPF">[none]</AR><AR Key="TREP" SN="TREP" LN="Transitional Epithelial" AF="0" NR="1
/LPF">[none]</AR><AR Key="REEP" SN="REEP" LN="Renal Epithelial" AF="0" NR="1
/LPF">[none]</AR><AR Key="OVFB" SN="OVFB" LN="Oval Fat Body" AF="0" NR="1
/HPF">[none]</AR><AR Key="FAT" SN="FAT" LN="Fat" AF="0" NR="1 /HPF">[none]</AR><AR

```

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Key="MUCS" SN="MUCS" LN="Mucous" AF="0" NR="MOD">[none]</AR><AR Key="RBCC" SN="RBCC" LN="Red Blood Cell Clump" AF="0" NR="1 /HPF">[none]</AR><AR Key="SPRM" SN="SPRM" LN="Sperm" AF="0" NR="3 /HPF">[none]</AR><AR Key="TRCH" SN="TRCH" LN="Trichomonas" AF="0" NR="1 /HPF">[none]</AR><AR Key="NSE" SN="NSE" LN="Non-Squamous Epithelial" AF="0" NR="1 /LPF">[none]</AR><AR Key="UNCC" SN="UNCC" LN="Unclassified Cast" AF="0" NR="1 /LPF">[none]</AR><AR Key="HYAL" SN="HYAL" LN="Hyaline Cast" AF="0" NR="2 /LPF">[none]</AR><AR Key="EPIC" SN="EPIC" LN="Epithelial Cast" AF="0" NR="1 /LPF">[none]</AR><AR Key="WBCT" SN="WBCT" LN="White Blood Cell Cast" AF="0" NR="1 /LPF">[none]</AR><AR Key="RBCT" SN="RBCT" LN="Red Blood Cell Cast" AF="0" NR="1 /LPF">[none]</AR><AR Key="GRAN" SN="GRAN" LN="Granular Cast" AF="0" NR="1 /LPF">[none]</AR><AR Key="CELL" SN="CELL" LN="Cellular Cast" AF="0" NR="1 /LPF">[none]</AR><AR Key="BROAD" SN="BROAD" LN="Broad Cast" AF="0" NR="1 /LPF">[none]</AR><AR Key="FATC" SN="FATC" LN="Fatty Cast" AF="0" NR="1 /LPF">[none]</AR><AR Key="WAXY" SN="WAXY" LN="Waxy Cast" AF="0" NR="1 /LPF">[none]</AR><AR Key="UNCX" SN="UNCX" LN="Unclassified Crystal" AF="0" NR="FEW">OCC</AR><AR Key="TPO4" SN="TPO4" LN="Triphosphate Crystal" AF="0" NR="FEW">[none]</AR><AR Key="CAOX" SN="CAOX" LN="Calcium Oxalate Crystal" AF="0" NR="FEW">[none]</AR><AR Key="CAPH" SN="CAPH" LN="Calcium Phosphate Crystal" AF="0" NR="FEW">[none]</AR><AR Key="CACB" SN="CACB" LN="Calcium Carbonate Crystal" AF="0" NR="FEW">[none]</AR><AR Key="URIC" SN="URIC" LN="Uric Acid Crystal" AF="0" NR="FEW">[none]</AR><AR Key="LEUC" SN="LEUC" LN="Leucine Crystal" AF="0" NR="POS">[none]</AR><AR Key="CYST" SN="CYST" LN="Cystine Crystal" AF="0" NR="POS">[none]</AR><AR Key="TYRO" SN="TYRO" LN="Tyrosine Crystal" AF="0" NR="POS">[none]</AR><AR Key="AMOR" SN="AMOR" LN="Amorphous Crystal" AF="0" NR="FEW">[none]</AR><AR Key="UNCL" SN="UNCL" LN="Unclassified" AF="0" NR="999999999 /HPF">[none]</AR></AC><CM>Your comment appears here!!!! Dr. JJ notified.</CM><ARV>42</ARV><AUD><XE OP="GregLoa" dt="2006-08-30 12:02:43"/><XE OP="[auto]" dt="2006-08-30 11:52:58"/></AUD><SysCfg MicroPresent="T" AXPPresent="T" iChem200Present="F"/></SA>

4.9. Example 9:

Upon completion of IRISpec CA™ Urine Chemistry Control, the following control results can be transmitted:

Data:

```
<?xml version="1.0" encoding="UTF-16"?>
<?xml-stylesheet type="text/xsl" href="C:\IRIS2K1\Templates\ChemQCResult-style.xsl"?>
<ChemQC NAME="CA" SID="AP" OP="GregLoa" ADT="2007-01-08 13:20:15-08:00" ADTS="2007-01-08 13:20:15" RDT="2007-01-09 08:27:00-08:00" RDTs="2007-01-09 08:27:00" LOT="CA" EXPIR="2004-06-01" DSID="AX-Stick" DSEXPIR="2004-06-01" RP="8" STATUS="P"><PF>Iris Diagnostics</PF><PF>9172 Eton Ave</PF><PF>Chatsworth, CA 91311</PF><PF>800.776.4747</PF><PF></PF><PF></PF><PF></PF><PF></PF><QCR Key="BLD" SN="BLD" LN="Blood" ISTATUS="P" LL="+1 (0.06 mg/dL)" UL="+3 (OVER)" UNIT="mg/dL">+2 (0.2 mg/dL)</QCR><QCR Key="BIL" SN="BIL" LN="Bilirubin" ISTATUS="P" LL="+1 (0.5 mg/dL)" UL="+3 (10 mg/dL)" UNIT="mg/dL">+1 (1 mg/dL)</QCR><QCR Key="URO" SN="URO" LN="Urobilinogen" ISTATUS="P" LL="Normal" UL="Normal" NIT="mg/dL">Normal</QCR><QCR Key="KET" SN="KET" LN="Ketone" ISTATUS="P" LL="-" UL="-" UNIT="mg/dL">-</QCR><QCR Key="GLU" SN="GLU" LN="Glucose" ISTATUS="P" LL="+- (30 mg/dL)" UL="+4 (OVER)" UNIT="mg/dL">+2 (200 mg/dL)</QCR><QCR Key="PRO" SN="PRO" LN="Protein" ISTATUS="P" LL="+- (10 mg/dL)" UL="+4 (OVER)" UNIT="mg/dL">+3 (600 mg/dL)</QCR><QCR Key="NIT" SN="NIT" LN="Nitrite" ISTATUS="P" LL="-" UL="-" UNIT="mg/dL">-</QCR><QCR Key="LEU" SN="LEU" LN="Leukocytes" ISTATUS="P" LL="Negative" UL="Negative" UNIT="Leu/ul">Negative</QCR><QCR Key="PH" SN="PH" LN="pH" ISTATUS="P" LL="7.5" UL="9.0" UNIT="mg/dL">8.5</QCR><QCR Key="SG" SN="SPGR" LN="Specific Gravity" ISTATUS="P" LL="1.000" UL="1.025" UNIT="mg/dL">1.012</QCR><QCR Key="COL" SN="COL" LN="Color" ISTATUS="P" LL="Green" UL="Green" UNIT="mg/dL">Green</QCR><QCR Key="CLA" SN="CLA" LN="Clarity" ISTATUS="P" LL="Clear" UL="Clear" UNIT="mg/dL">Clear</QCR></ChemQC>
```

4.10. Example 10:

Upon completion of the iQ® Positive Control, the following control results can be transmitted:

Data:

```
<?xml version="1.0" encoding="UTF-16"?><cr><lf><?xml-stylesheet type="text/xsl"
href="C:\IRIS2K1\Templates\CountControlResult-style.xsl"?><cr><lf><CQC CCTYPE="+L"
ID="PC04184H1005000000" SID="AP" OP="q" ADT="2004-10-13 09:15:20-08:00" ADTS="2004-10-13
09:15:20" RDT="2004-11-19 10:07:07-08:00" RDTs="2004-11-19 10:07:07" LOT="184-2004-PC"
EXPIR="03-2005" PRODID="PC" MEAN="1005" MNC="0" RP="6" SQN="0"
RSQN="29"><PF>IRIS</PF><PF>Chatsworth, California
</PF><PF></PF><PF></PF><PF></PF><PF></PF><PF></PF><PF></PF><PF></PF><MIN>804</MIN><MAX>1206</MAX>
<COUNT>921</COUNT><STATUS>P</STATUS><REF>2.265</REF></CQC><cr><lf>
```

4.11. Example 11:

Upon completion of a Body Fluid Control, the following Body Fluid Control results can be transmitted:

Data:

```
<?xml version="1.0" encoding="UTF-16"?>
<?xml-stylesheet type="text/xsl" href="C:\IRIS2K1\Templates\BFControlResult-style.xsl"?>
<BFQC BFCTYPE="BL2" ID="L206205E4810272000" SID="AP" OP="Manager" ADT="2006-
08-24 15:37:12-08:00" ADTS="2006-08-24 15:37:12" RDT="2006-08-30 11:35:27-08:00"
RDTs="2006-08-30 11:35:27" LOT="205-2006-L2" EXPIR="12-2006" PRODID="L2"
RBCMEAN="48100" NUCMEAN="2700" RP="3" SQN="0" RSQN="24"><PF>Iris LIS
Laboratory</PF><PF>9172 Eton Ave</PF><PF>Chatsworth, CA
91311</PF><PF>800.776.4747</PF><PF></PF><PF></PF><PF></PF><PF></PF><RBCMIN>33670<
/RBCMIN><RBCMAX>62530</RBCMAX><RBCCOUNT>29831</RBCCOUNT><NUCMIN>
1889</NUCMIN><NUCMAX>3510</NUCMAX><NUCCOUNT>2496</NUCCOUNT><STAT
US>F</STATUS><REF>2.408</REF></BFQC>
```

4.12. Example 12:

Upon completion of a body fluid analysis, the following body fluid results can be transmitted:

Data:

```
<?xml version="1.0" encoding="UTF-16"?><cr><lf><?xml-stylesheet type="text/xsl"
href="C:\IRIS2K1\Templates\SpecimenAnalysis-style.xsl"?><cr><lf><SA BF="PER"
ID="023400858" SID="AP" OP="gloa" ADT="2005-06-10 15:06:54-08:00" ADTS="2005-06-
10 15:06:54" RDT="2005-06-10 17:12:59-08:00" RDTs="2005-06-10 17:12:59" RP="3"
SQN="208" RSQN="25" DILN="1" DILD="20" IMP="0" REDT="2005-06-10 15:14:44-08:00"
REDTS="2005-06-10
15:14:44"><PF></PF><PF></PF><PF></PF><PF></PF><PF></PF><PF></PF><PF></PF><PF></PF><PF></PF>
<AC AT="Sediment" AS="Done" SO="Internal"><AR Key="RBC" SN="Red Blood Cells"
LN="Red Blood Cells" AF="0" NR="99999999 /uL">14666 /uL</AR><AR Key="NUCL"
SN="Nucleated Cells" LN="Nucleated Cells" AF="0" NR="99999999 /uL">15982
/uL</AR><AR Key="UNCX" SN="Crystals" LN="Crystals" AF="0"
```

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NR="Observed">Observed</AR><AR Key="BACT" SN="Bacteria" LN="Bacteria" AF="0"
 NR="Observed">Observed</AR></AC><CM>Sample is contaminated!!!! Dr. Smith is
 notified 01/01/01 17:55 !!!!!</CM><ARV>42</ARV></SA><cr></f>

5. Appendix

Data Format	ASCII encoded Unicode XML
Interface Mode	Unidirectional; Bidirectional Host Query
Protocol	ASTM-E 1381-95 low level; IRIS-defined XML high level
Communication (Default)	Baud rate: 9600 Bata Bits: 8 Stop Bit: 1 Parity: None Flow Control: None
Upload Codes	Chemistry for Urine: BLD = Blood; BIL = Bilirubin; URO = Urobilinogen; KET = Ketone; GLU = Glucose; PRO = Protein; NIT = Nitrite; LEU = Leukocytes; PH = pH; SG = Specific Gravity; COL = Color; CLA = Clarity. Microscopy for Urine: ART = Artifact; RBC = Red Blood Cell; DRBC = Dysmorphic Red Blood Cell WBC = White Blood Cell; WBCC = White Blood Cell Clump; BACT = Bacteria; BYST = Budding Yeast; HYST = Hyphae Yeast; SQEP = Squamous Epithelial; TREP = Transitional Epithelial; REEP = Renal Epithelial; OVFB = Oval Fat Body; FAT = Fat; MUCS = Mucous; RBCC = Red Blood Cell Clump; SPRM = Sperm; TRCH = Trichomonas; NSE = Non-Squamous Epithelial; UNCC = Unclassified Cast; HYAL = Hyaline Cast; EPIC = Epithelial Cast; WBCT = White Blood Cell Cast; RBCT = Red Blood Cell Cast; GRAN = Granular Cast; CELL = Cellular Cast; BROAD = Broad Cast; FATC = Fatty Cast; WAXY = Waxy Cast; UNCX = Unclassified Crystal; TPO4 = Triple Phosphate Crystal; CAOX = Calcium Oxalate; CAPH = Calcium Phosphate Crystal; CACB = Calcium Carbonate Crystal; URIC = Uric Acid Crystal; LEUC = Leucine Crystal; CYST = Cystine Crystal; TYRO = Tyrosine Crystal; AMOR = Amorphous Crystal, UNCL = Unclassified; PC = All Small Particles, if instrument is configured to report All Small Particles. Microscopy for Body Fluid: RBC = Red Blood Cells, NUCL = Nucleated Cells, UNCX = Crystals, BACT = Bacteria
Pin Assignments	Pin 2 = Output Pin 3 = Input Pin 5 = Ground
Data Port	DB9M

REVISIONS	WRITTEN/REVISED BY	COMMENTS
A	Greg Loa	
B	Greg Loa	
C	Greg Loa	
D	Greg Loa	
E	Carol Eden	Updated Header and Footer
F	Greg Loa	Updated transmission samples and screenshots. Added DRBC in the Appendix.